Arijit Roy

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Education

East Delta University Sept 2022 – Sept 2026
Bachelor of Science (B.Sc.) in Computer Science and Engineering **CGPA: 3.72 / 4.00**
*Chairman's Distinguished Scholarship; Dean's List **6×**; Vice-Chancellor's List **1×**.*

Publication

FedAdapt-EF: An Efficient Federated Learning-Based Approach on Adaptive Compression with Error Feedback for Diabetic Retinopathy

Published – 2026 International Conference on Circuit, Systems and Communication (ICCSC)

Proposed a federated learning framework for diabetic retinopathy screening that reduces communication bandwidth by **23.9%** without sacrificing accuracy, enabling AI-assisted diagnosis in low-bandwidth rural clinics. Achieved **AUC 0.9601** across 16,652 retinal images, outperforming the uncompressed baseline.

Projects

ML Study Buddy [GitHub] — Next.js, FastAPI, Python, FAISS, Groq LLM — [Live]

- Built a Retrieval-Augmented Generation (RAG) system with Next.js frontend and FastAPI backend for AI-powered machine learning tutoring with document retrieval.
- Fine-tuned sentence-transformer embeddings via contrastive learning, achieving **15–25% improvement** in retrieval metrics (Recall@5, MRR, NDCG).
- Integrated Whisper STT, SpeechT5 TTS, PDF ingestion, and OCR (TrOCR) for multimodal interaction via FAISS vector store and Groq LLM API.

FloodGuard Aid Network [GitHub] — React, Node.js, MongoDB, Express.js — [Live]

- Built a full-stack MERN disaster relief platform connecting flood victims with volunteers; won recognition at a national hackathon organized by Programming Hero.
- Implemented geolocation, mapping, and image upload for real-time aid coordination via RESTful API with Express.js and MongoDB.

AcademiaConnect [GitHub] — MongoDB, Express.js, React, Node.js (MERN Stack)

- Designed a university social platform with posts, comments, friend connections, and dynamic class schedules using OOP-based modular architecture.
- Implemented JWT-based authentication, full profile management, and RESTful backend API with Express.js and MongoDB.

AcademiaConnectApp [GitHub] — React Native, Supabase

- Developed a cross-platform mobile app for university students with real-time notifications, friend requests, and schedule viewer.

Extracurricular Activities

Coordinator, EDU Computer Club 2025 – Present

- Led activities for **250+ members**; organized workshops, contests, and guest sessions; planned and executed **EDU HackFest 2025** end-to-end.
- Secured **∼100,000 BDT** in sponsorships; managed cross-functional teams and external organization partnerships.

Competitive Programmer 2023 – Present

- Active participant in inter-university contests: **ICPC Asia Dhaka Regional 2024**, BUET CSE Fest 2025, UIU CSE Fest 2024.

Skills

- **Programming:** C++, Python, JavaScript, SQL
- **Web & App:** React, Next.js, React Native, Node.js, Express.js, REST APIs, HTML/CSS, Tailwind CSS
- **Databases:** MongoDB, MySQL, Supabase, PostgreSQL
- **AI / ML:** Machine Learning, Deep Learning, RAG, Sentence Transformers, FAISS, Federated Learning
- **Concepts:** Object-Oriented Programming (OOP), Data Structures & Algorithms, RESTful API Design, STL, Multi-threading
- **Languages:** Bangla (Native), English (Proficient)